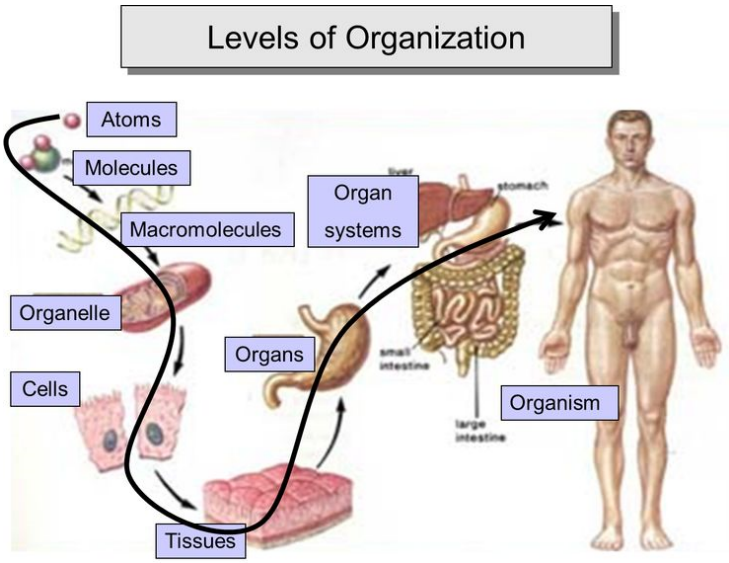
Levels of Structural Organization in the Human Body

The human body has 6 main levels of structural organization. We will begin this lesson with the simplest level within the structural hierarchy.



Chemical level– is the simplest level within the structural hierarchy. The chemical level includes the tiniest building blocks of matter, **atoms**, which combine to form **molecules**, like water. In turn, molecules combine to form **organelles**, the internal organs of a cell.

Cellular level– the cellular level is made up of the smallest unit of living matter, the **cell**. Individual cells may have some common functions but vary widely in size and shape. Each type of cells carries out a set of unique tasks within the human body.

Tissue level– Tissues are groups of similar cells that have a common function. A **tissue** must contain two different types of cells. The four basic tissue types in humans include **epithelium**, **connective**, **muscle**, and **nervous** tissue. Each tissue has a characteristic role within the human body which we will discuss later.

Organ level– an **organ** is a structure composed of at least two different tissue types that perform a specific function within the body. Examples include the brain, stomach, and liver. Complex functions begin to emerge at this level.

Levels of structural organization that make up the human body.

Organ system level– One or more organs work in unison to accomplish a common purpose. For instance, the heart and blood vessels work together and circulate blood throughout the body to provide oxygen and nutrients to cells. Besides the cardiovascular system, the other organ systems of the body are the integumentary, skeletal, nervous, muscular, endocrine, respiratory, lymphatic, digestive, urinary, and reproductive systems.

Organismal level– The organismal level is the highest level of organization. It is the sum total of all structural levels working together. In short, it is the human being (or organism) as a whole.

Review

- Name the six different levels of structural organization in the human body and explain their relationships.
- List the 11 organ systems of the human body and explain their major functions.

Check your understanding. Answer the questions below using complete sentences.

1. In the hierarchy, at which level of structural organization would an immunologist's field of study be considered? Why?

2. What is the correct structural order of the following terms: atom, organ, cell, organism, tissue?

3. Which organ system includes the heart and blood vessels? Which organ system includes the kidneys?

4. List the order of the **levels of organization** in biological life forms **from simplest to most complex** and give an **example** of something from each level of the scale.

Word Bank: tissue, organelle, organ, cell, organism, organ system

Biological Level of Organization	Example